(3)(i) $x = (2n+1)\frac{\pi}{6}$ or $x = \frac{n\pi}{2} + (-1)^n \frac{\pi}{12}$, $n \in \mathbb{Z}$

(ii) $\theta = n\pi + (-1)^n \frac{\pi}{6}$ or $\theta = n\pi + (-1)^n \frac{\pi}{2}$ $n \in \mathbb{Z}$

(iii) $\theta = (2n+1)\frac{\pi}{4}$ or $\theta = 2n\pi$, $n \in \mathbb{Z}$

(iv) $\theta = \frac{n\pi}{3}$ or $\theta = n\pi \pm \frac{\pi}{3}$, $n \in \mathbb{Z}$

(v) $\theta = 2n\pi$ or $\theta = \frac{2n\pi}{3} + \left(\frac{-\pi}{6}\right)$, $n \in \mathbb{Z}$

(vi) $\theta = (8n+1)\frac{\pi}{4}$, $n \in \mathbb{Z}$

(vii) $\theta = 2n\pi + \frac{\pi}{6} \pm \frac{\pi}{3}$, $n \in \mathbb{Z}$

(viii) $\theta = 2n\pi - \frac{\pi}{2} \pm \frac{2\pi}{3}$, $n \in \mathbb{Z}$

(ix) $\theta = \frac{n\pi}{3} + \frac{\pi}{18}$, $n \in \mathbb{Z}$

(x) $\theta = n\pi \pm \frac{\pi}{10}$, $n \in \mathbb{Z}$

(xi) $x = 2n\pi \pm \frac{\pi}{3}$ $n \in \mathbb{Z}$

Exercise 3.9

(2) $\angle A = 75°$ (9) $40m, 40m, 40m$ (10) $4m,\ 4m,\ 4m$ and $4\sqrt{3}\ sq.meter.$

Exercise 3.10

(1) no such triangle exist (3) $\angle A = 15°, \angle B = 105°$ (7) $\frac{5\sqrt{2}}{\sqrt{3}-1}$ km (8) $2\sqrt{13}$ km

(9) $3 + \sqrt{73}$ km (10) 7 km

(11) Total Cost: 155800 and Perimeter $180 + 20\sqrt{27}$feet

(12) $x = 100$ km (13) $\sqrt{5 - 2\sqrt{2}}$ km

(14) $2\sqrt{6} + 2\sqrt{3} + 6$ km (15) $AB = 10\sqrt{3}$ km

Exercise 3.11

(1)(i) $\theta = \frac{\pi}{4}$ (ii) $\theta = \frac{\pi}{6}$ (iii) $\theta = -\frac{\pi}{2}$ (iv) $\theta = \frac{3\pi}{4}$ (v) $\theta = \frac{\pi}{3}$

Exercise 3.12

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
(4)	(1)	(1)	(1)	(4)	(4)	(1)	(2)	(4)	(2)	(3)	(2)	(3)	(3)	(2)

16	17	18	19	20
(4)	(1)	(1)	(1)	(1)